Fall 2026 DSC 242: Homework 6

Due 11:59PM Friday, November 13

Submission. Part 3(b) is experimental: submit it as a PDF report (for example, exported from a Jupyter notebook) that contains your figure and table, a short discussion, and the code that produced them (as notebook cells or as an appendix). The figure should have labeled axes, a legend, and a caption stating the parameters and the number of trials. The other parts are written problems.

1. (25 points) Let $X_1, \dots, X_n$ be i.i.d. random points uniformly distributed in the unit square $[0,1]^2$.

   (a) (10 points) Show that for an absolute constant $C > 0$ and any $n \geq 2$,

   $$\mathbb{E}\Big[\min_{2 \leq i \leq n} \|X_1 - X_i\|_2^2\Big] \leq \frac{C}{n}.$$

   (b) (15 points) Let $f(X_1, \dots, X_n)$ be the total length of the shortest path that visits each point exactly once and returns to the starting point. (This is known as the solution to the *traveling salesman problem*.) Show that for an absolute constant $C > 0$ and any $n \geq 2$,

   $$\operatorname{Var} f(X_1, \dots, X_n) \leq C.$$

2. (van Handel 2.9, 25 points) Let $X_1, \dots, X_n$ be i.i.d. standard exponential random variables (having density e^{-x} over positive values $x > 0$) and let $f : \mathbb{R}^n \to \mathbb{R}$ be any bounded function with bounded first partial derivatives. Show the Poincaré inequality

   $$\operatorname{Var} f(X_1, \dots, X_n) \leq 4\mathbb{E}\|\nabla f(X_1, \dots, X_n)\|_2^2.$$

   Hint: Use the identity $\int_0^\infty g(x)e^{-x}\mathrm{d}x = g(0) + \int_0^\infty g'(x)e^{-x}\mathrm{d}x$.

3. (Order statistics, 30 points) Let $X_1, \dots, X_n$ be any independent random variables, and let $X_{(1)} \geq X_{(2)} \geq \cdots \geq X_{(n)}$ be their ordered values.

   (a) (BLM 3.3, 20 points) Show that for any $1 \leq k \leq n-1$,

   $$\operatorname{Var}[X_{(k)}] \leq k\mathbb{E}[(X_{(k)} - X_{(k+1)})^2].$$

   (b) (Experiment: how tight are the variance bounds?, 10 points) Let $X_1, \dots, X_n$ be i.i.d., either (G) $\mathcal{N}(0,1)$ or (E) Exp(1). We compare three upper bounds on $\operatorname{Var} X_{(k)}$.

   - The bound $k\mathbb{E}[(X_{(k)} - X_{(k+1)})^2]$ of part (a).
   - The Poincaré bound. The function $x \mapsto x_{(k)}$ (the k-th largest coordinate of $x \in \mathbb{R}^n$) satisfies $\|\nabla x_{(k)}\|_2 = 1$ for almost every x (check this), so the Gaussian Poincaré inequality gives $\operatorname{Var} X_{(k)} \leq 1$ for (G), and Problem 2 gives $\operatorname{Var} X_{(k)} \leq 4$ for (E) (Problem 2 extends from bounded f to $f(x) = x_{(k)}$ by truncation, which you may take for granted).

- The jackknife. For a statistic S defined for every sample size, let $S_{(i)}$ be S computed from the sample with X_i removed, and $\bar{S} = \frac{1}{n}\sum_{i=1}^{n} S_{(i)}$. The *jackknife estimate* of $\operatorname{Var} S(X_1, \dots, X_n)$ is

$$\widehat{V}_J = \frac{n-1}{n} \sum_{i=1}^{n} \big(S_{(i)} - \bar{S}\big)^2.$$

  Efron and Stein (1981) showed that it overestimates on average:

$$\frac{n}{n-1}\mathbb{E}\widehat{V}_J \geq \operatorname{Var} S(X_1, \dots, X_{n-1}) \qquad \text{for every symmetric } S;$$

  this is where the Efron–Stein inequality comes from. For $S =$ the k-th largest value (k fixed),

$$\widehat{V}_J = \Big(1 - \frac{1}{n}\Big)\Big(1 - \frac{k}{n}\Big)\, k\, \big(X_{(k)} - X_{(k+1)}\big)^2$$

  (check this), so it can be computed from the ordered sample. Unlike the bound of part (a), $\widehat{V}_J$ is computed from a single sample, without knowing the distribution.

For $n \in \{10, 10^2, 10^3, 10^4\}$ and $k \in \{1, \frac{n}{2}\}$ (the maximum and a median), estimate by Monte Carlo, with at least 2000 trials, the four quantities

$$\operatorname{Var} X_{(k)}, \qquad k\mathbb{E}[(X_{(k)} - X_{(k+1)})^2], \qquad \mathbb{E}\widehat{V}_J, \qquad \frac{\operatorname{sd}(\widehat{V}_J)}{\mathbb{E}\widehat{V}_J},$$

the last one being the relative standard deviation of $\widehat{V}_J$. (`numpy.partition` finds $X_{(k)}$ and $X_{(k+1)}$ without sorting the whole sample.) Plot the first three against n on log-log axes, one panel per distribution, one color per k, together with the Poincaré bound (format: Figure 1), and list all four in a table (figure and table: 4 points). Then discuss:

i. (2 points) How does $\operatorname{Var} X_{(k)}$ depend on n in each of the four cases, and which of the bounds follow this dependence? Explain why no bound of the form $C\,\mathbb{E}\|\nabla f(X)\|_2^2$ can do so for $f(x) = x_{(k)}$.

ii. (2 points) At $n = 10^4$, by what factor does the bound of part (a) exceed $\operatorname{Var} X_{(k)}$ for $k = 1$, and for $k = \frac{n}{2}$? By what factor does $\mathbb{E}\widehat{V}_J$ exceed it?

iii. (2 points) Does $\operatorname{sd}(\widehat{V}_J)/\mathbb{E}\widehat{V}_J$ go to 0 as n grows? For comparison, for the sample mean $S = \frac{1}{n}\sum_i X_i$ one can check that $\widehat{V}_J = s^2/n$, where s^2 is the sample variance, so $\widehat{V}_J / \operatorname{Var} S \to 1$ in probability. Would you use the jackknife to estimate the variance of a sample median, e.g., to build a confidence interval?

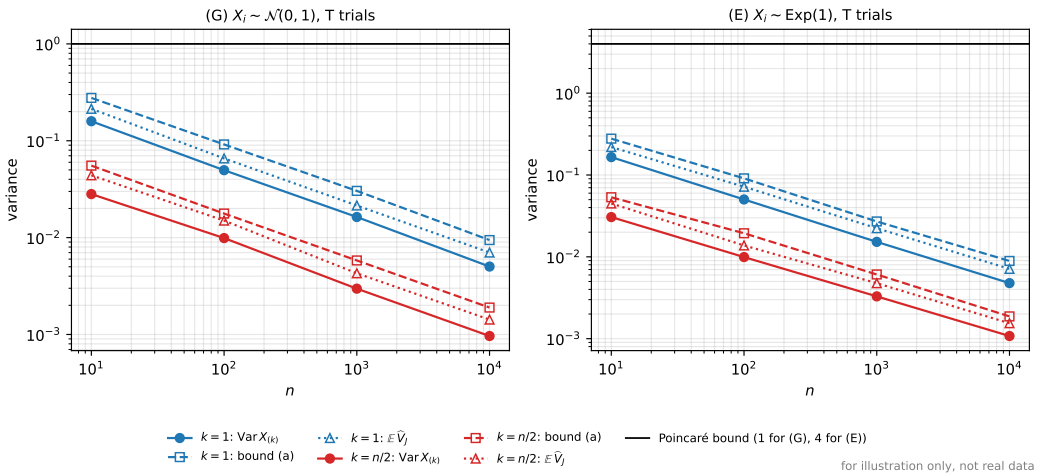


Figure 1: Expected format of the figure for part 3(b), for illustration only (not real data).

4. (van Handel 3.12, 20 points) For a positive random variable Y, let $\mathrm{Ent}(Y) = \mathbb{E}[Y \log Y] - \mathbb{E}[Y] \log \mathbb{E}[Y]$. By the Herbst argument (Lecture 14), if a random variable X satisfies, for some $\sigma^2 > 0$,

$$\mathrm{Ent}(e^{\lambda X}) \le \frac{\lambda^2 \sigma^2}{2} \mathbb{E}[e^{\lambda X}] \qquad \text{for all } \lambda \in \mathbb{R},$$

then $X - \mathbb{E}X$ is σ^2-sub-Gaussian (apply the Herbst argument to X and to $-X$). Show the following converse implication: if $X - \mathbb{E}X$ is $\frac{\sigma^2}{4}$-sub-Gaussian, then the above entropy inequality holds. We may therefore view this property as yet another equivalent formulation of the sub-Gaussian property.